

# What do neural networks actually learn?

Generalisation as a window into learning

Wieland Brendel

Empirical Theory in Representation Learning  
ECCV 2026

## An experimental perspective

# Psychophysics has a long history of concise probing experiments to uncover causal mechanisms



Original



Sample 1



Sample 2

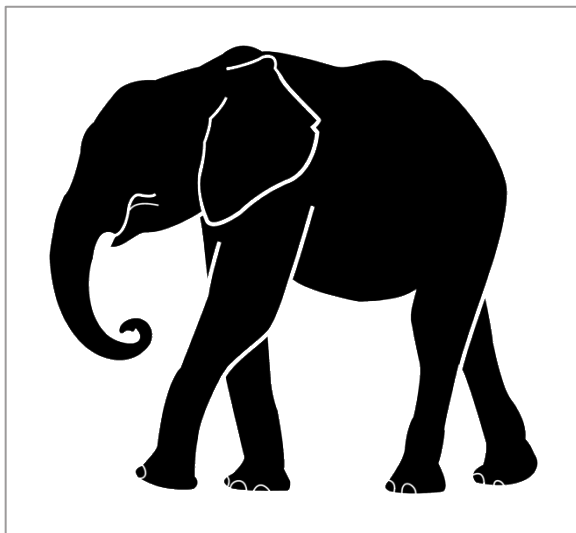
### Metamers of the ventral stream

J. Freeman, E. P. Simoncelli

Nature Neuroscience, 2011

# Are machines using shape or texture?

SHAPE



TEXTURE



**ImageNet-trained CNNs are biased towards texture; increasing shape bias improves accuracy and robustness**

R. Geirhos, P. Rubisch, C. Michaelis, M. Bethge, F. A. Wichmann, W. Brendel

ICLR 2019

# Machines use texture, not shape

**CAT**  
*SHAPE*



**ELEPHANT**  
*TEXTURE*

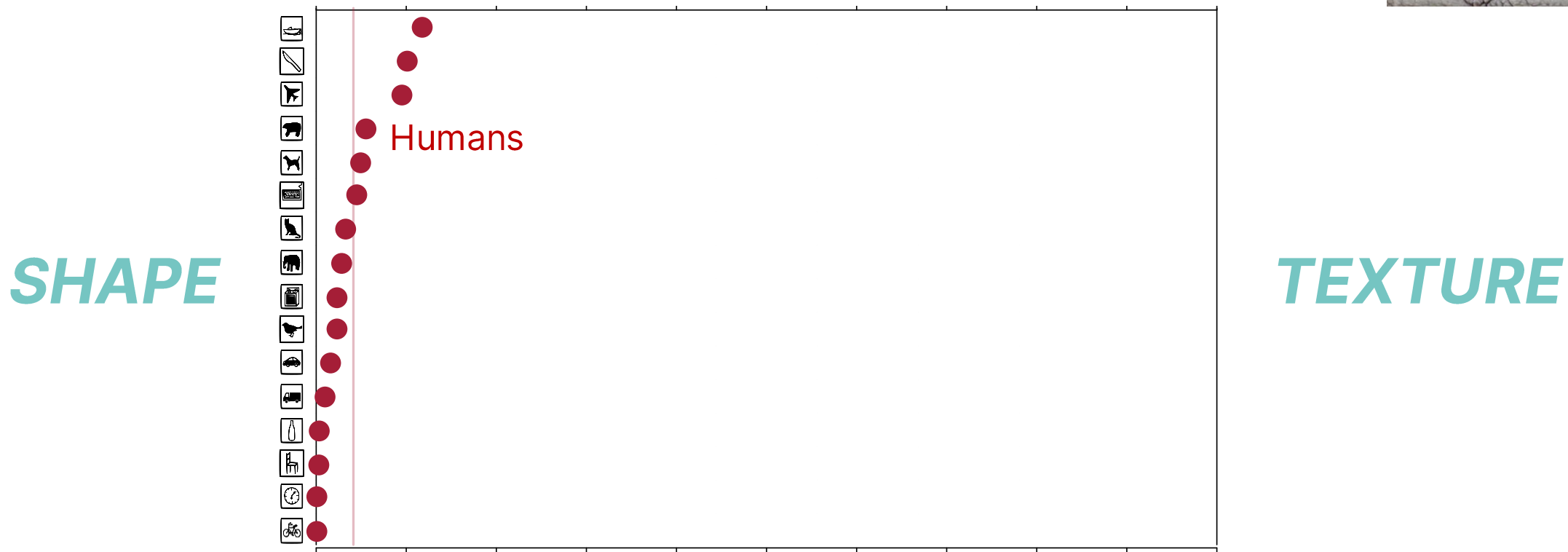


**ImageNet-trained CNNs are biased towards texture; increasing shape bias improves accuracy and robustness**

R. Geirhos, P. Rubisch, C. Michaelis, M. Bethge, F. A. Wichmann, W. Brendel

ICLR 2019

# Machines use texture, not shape



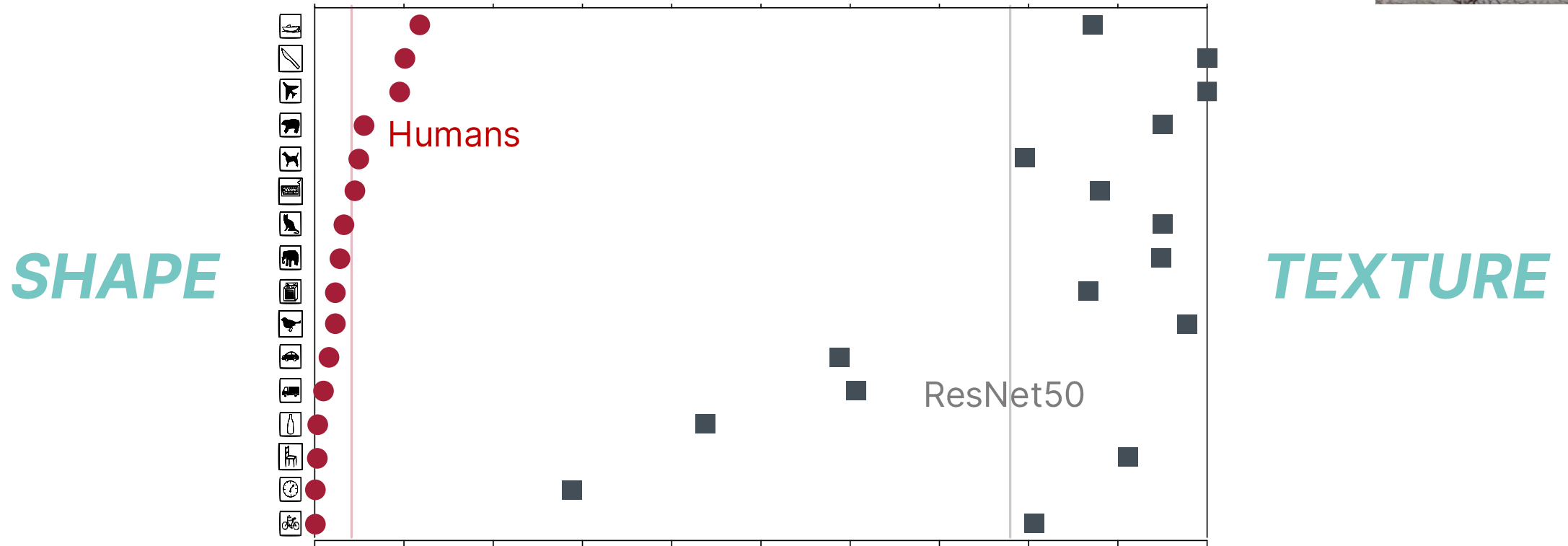
## ImageNet-trained CNNs are biased towards texture; increasing shape bias improves accuracy and robustness

R. Geirhos, P. Rubisch, C. Michaelis, M. Bethge, F. A. Wichmann, W. Brendel

ICLR 2019

## Diagnosing shortcut learning

# Machines use texture, not shape



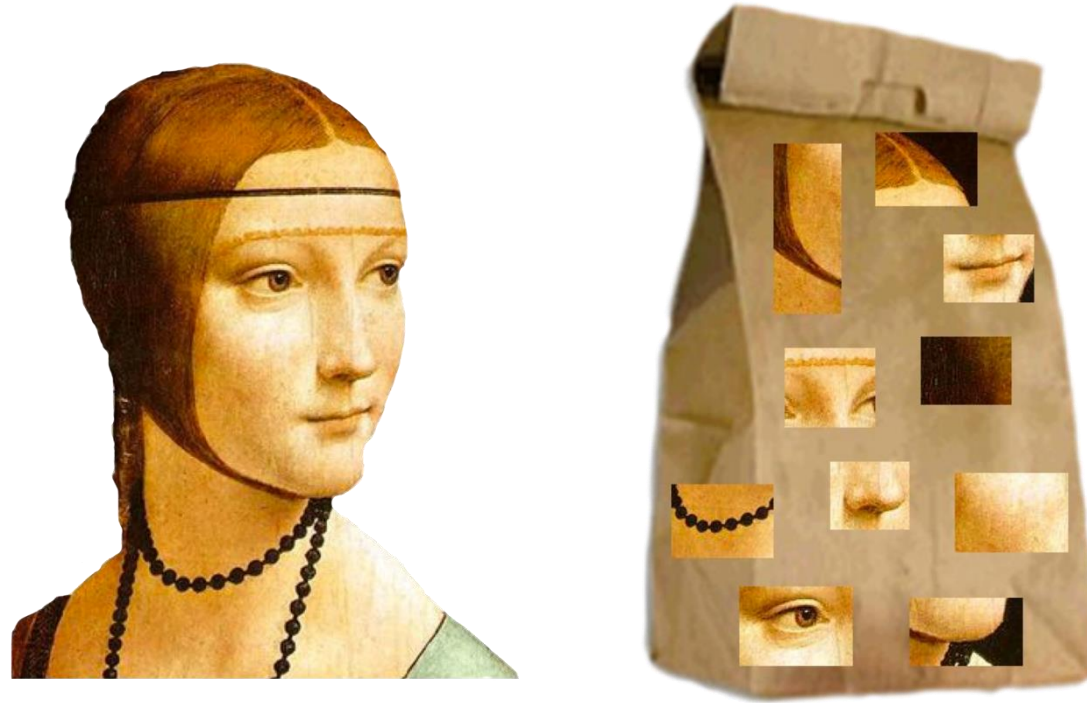
ImageNet-trained CNNs are biased towards texture; increasing shape bias improves accuracy and robustness

R. Geirhos, P. Rubisch, C. Michaelis, M. Bethge, F. A. Wichmann, W. Brendel

ICLR 2019

Early experiments · architectural test of the texture hypothesis

# CNNs ignore object composition



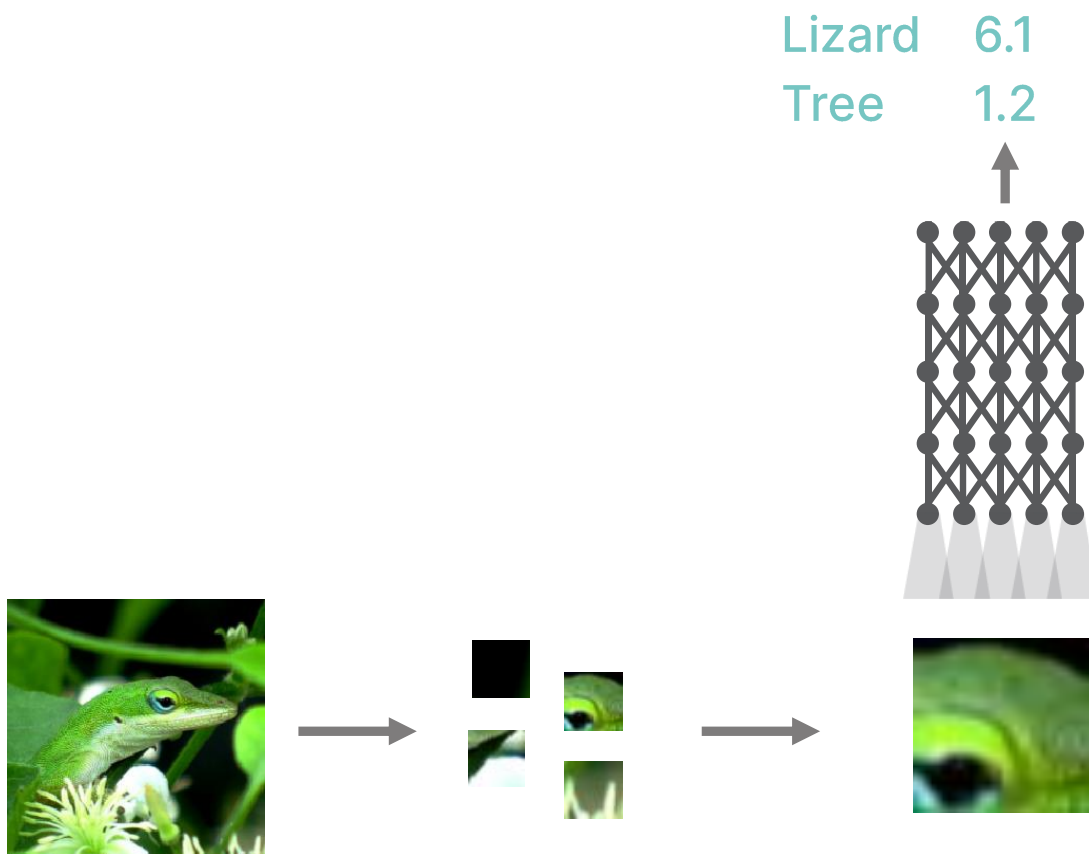
**Approximating CNNs with bag-of-local-features models works surprisingly well on ImageNet**

W. Brendel, M. Bethge

ICLR 2019

Early experiments · architectural test of the texture hypothesis

# CNNs ignore object composition

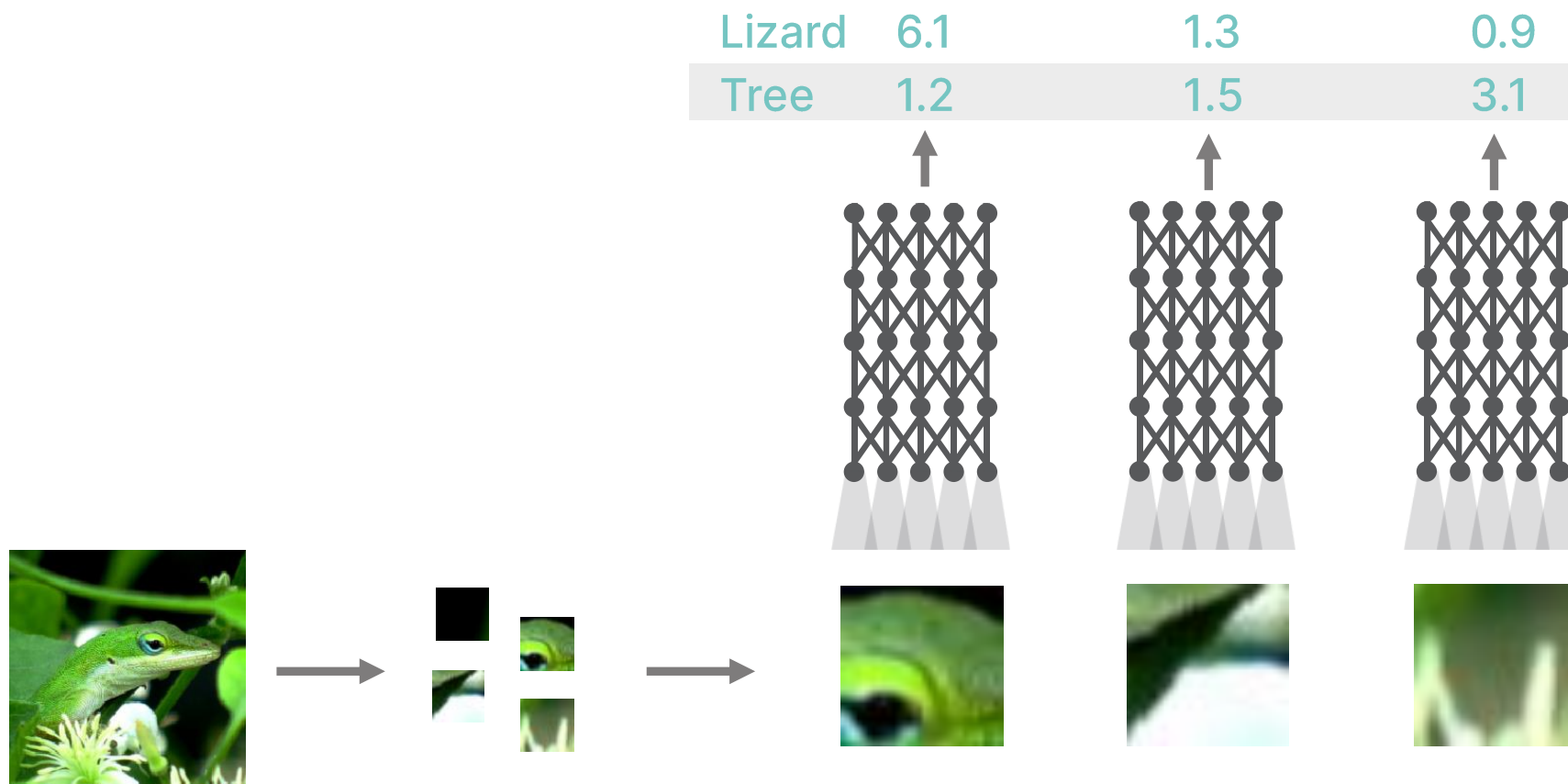


**Approximating CNNs with bag-of-local-features models works surprisingly well on ImageNet**

W. Brendel, M. Bethge

ICLR 2019

# CNNs ignore object composition

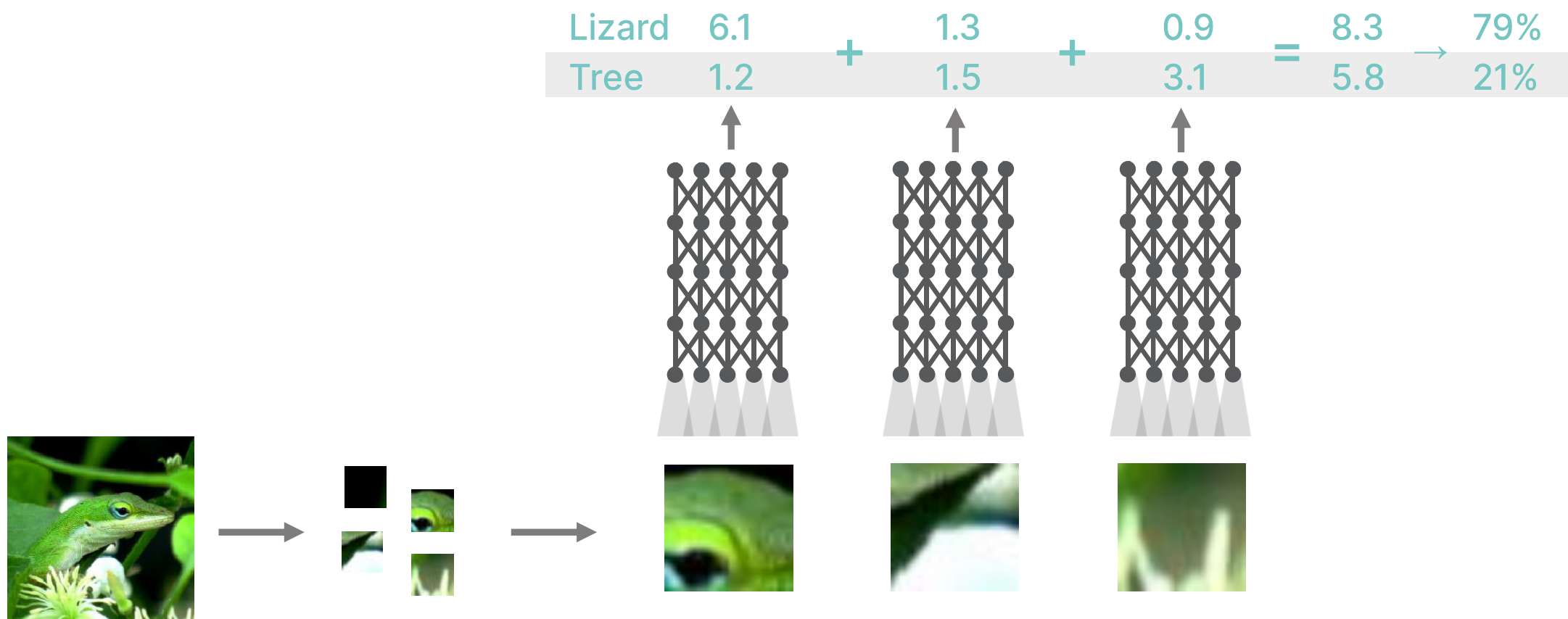


Approximating CNNs with bag-of-local-features models works surprisingly well on ImageNet

W. Brendel, M. Bethge

ICLR 2019

# CNNs ignore object composition



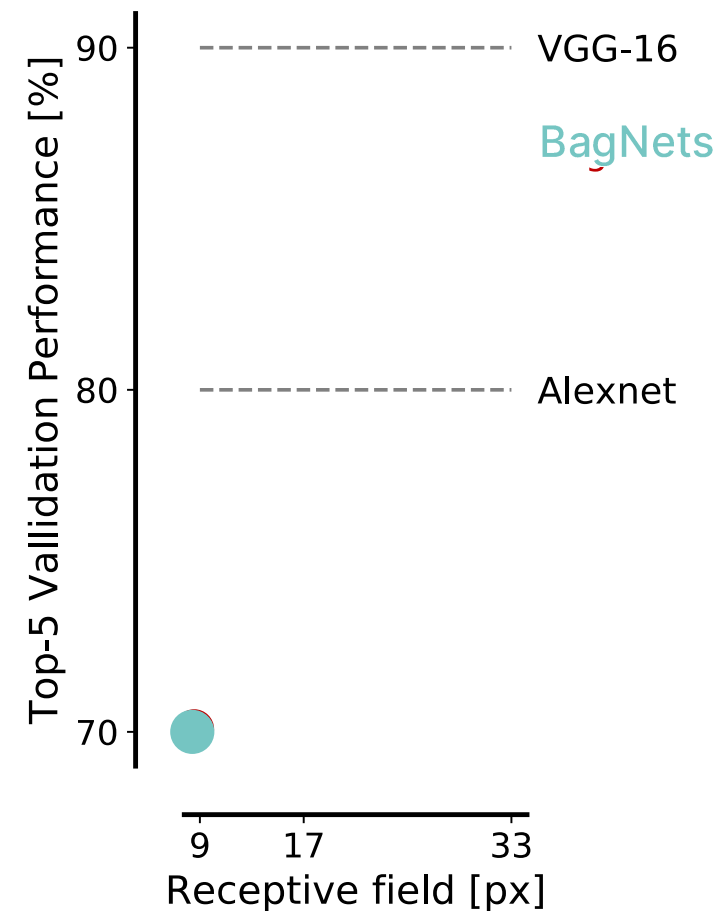
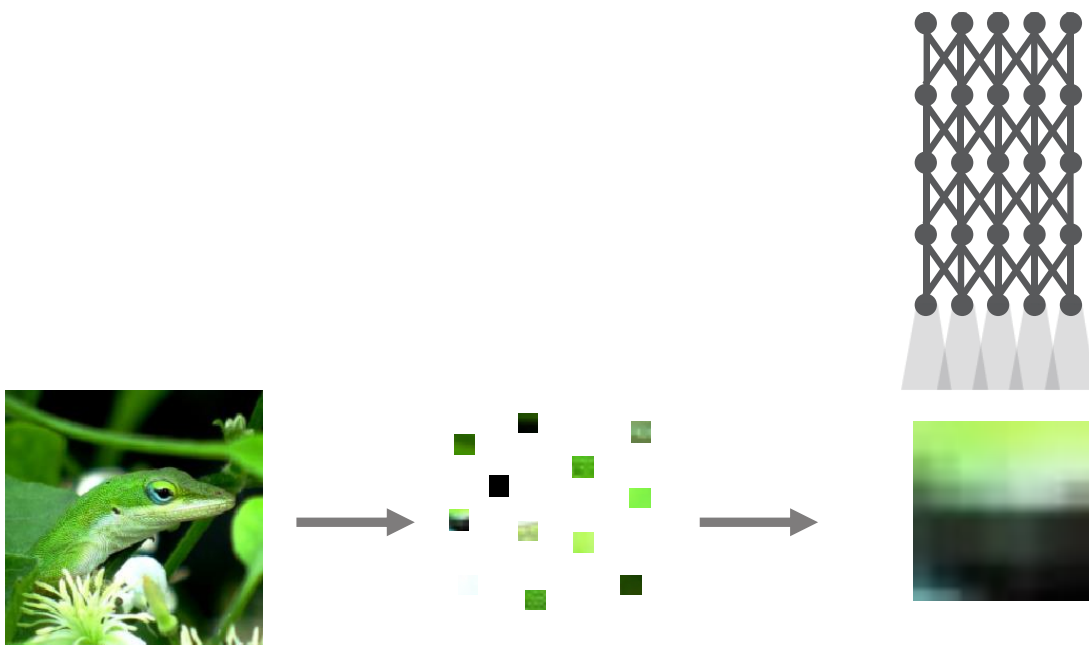
Approximating CNNs with bag-of-local-features models works surprisingly well on ImageNet

W. Brendel, M. Bethge

ICLR 2019

Early experiments · architectural test of the texture hypothesis

# CNNs ignore object composition



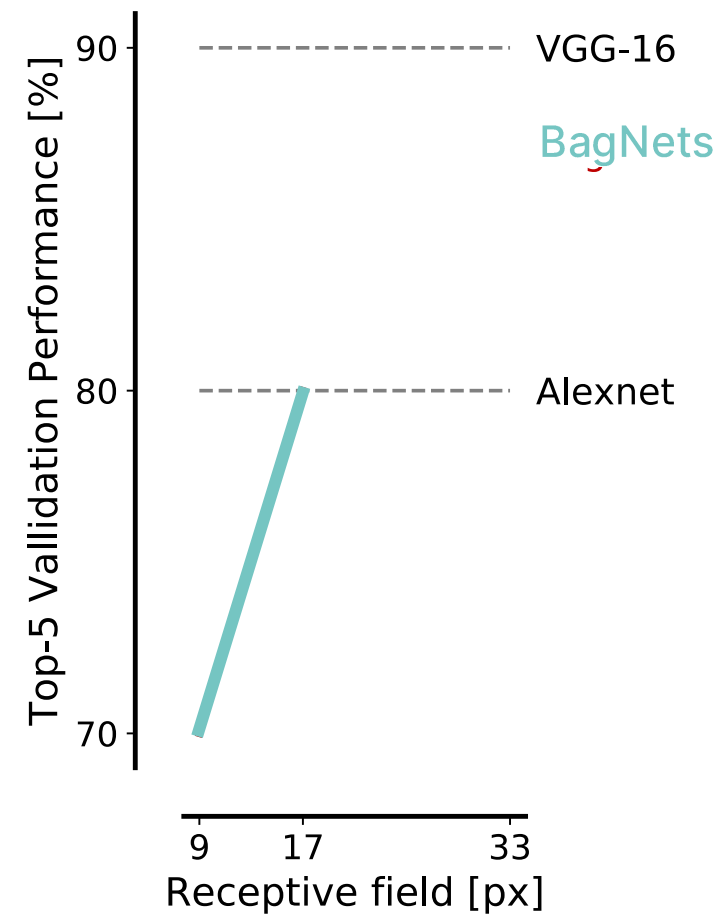
Approximating CNNs with bag-of-local-features models works surprisingly well on ImageNet

W. Brendel, M. Bethge

ICLR 2019

Early experiments · architectural test of the texture hypothesis

# CNNs ignore object composition



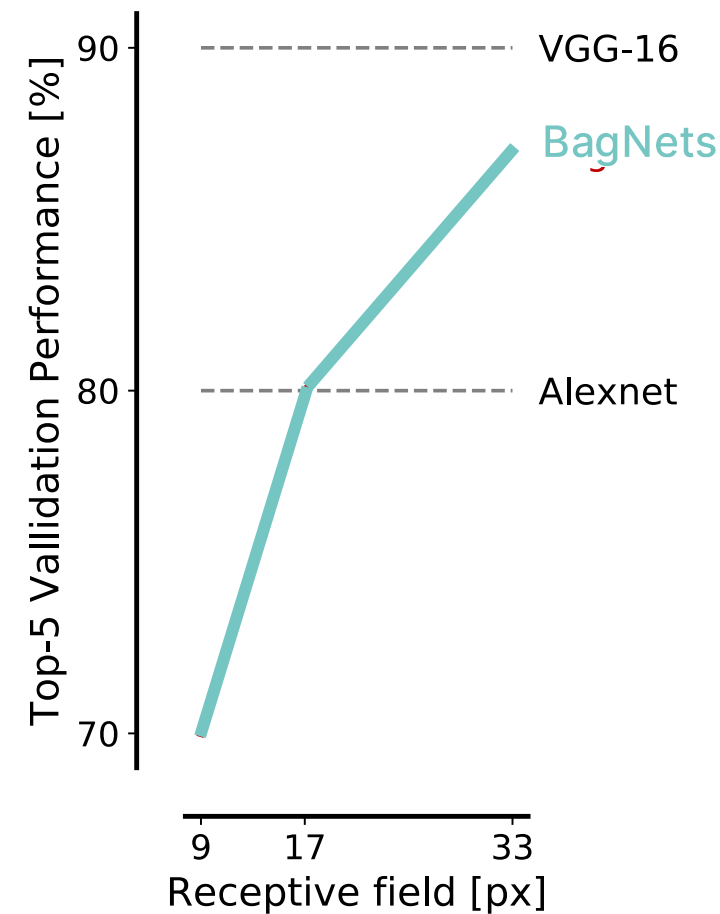
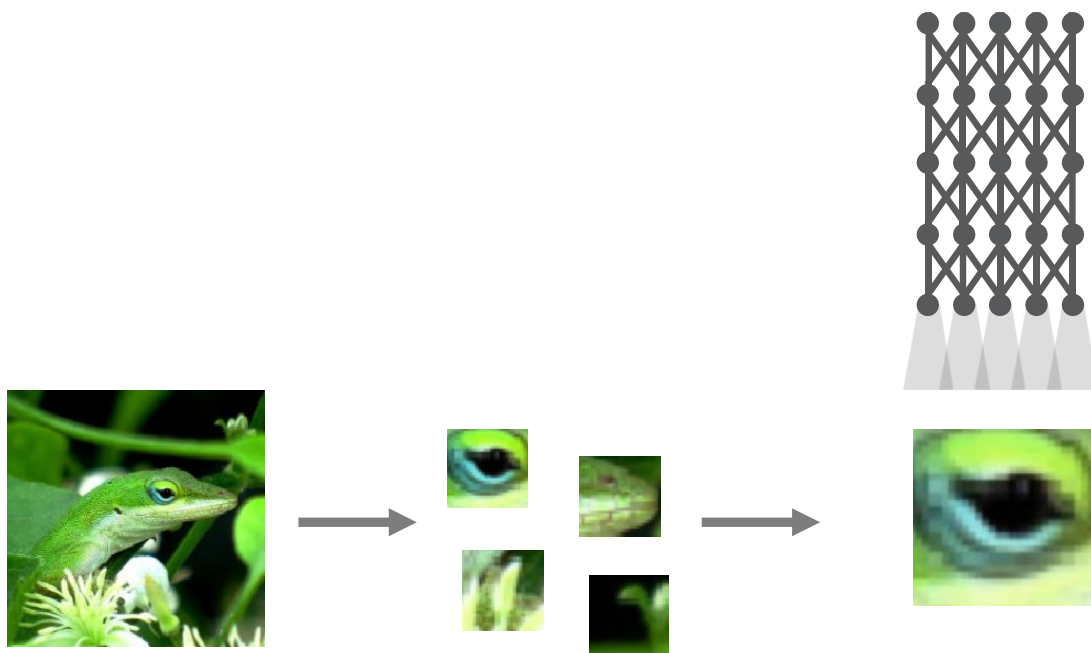
Approximating CNNs with bag-of-local-features models works surprisingly well on ImageNet

W. Brendel, M. Bethge

ICLR 2019

Early experiments · architectural test of the texture hypothesis

# CNNs ignore object composition



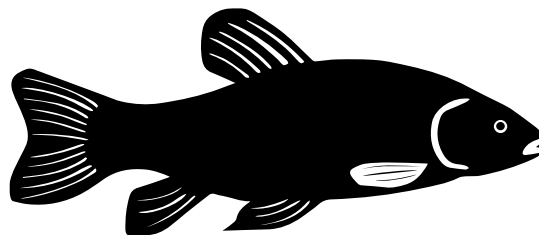
Approximating CNNs with bag-of-local-features models works surprisingly well on ImageNet

W. Brendel, M. Bethge

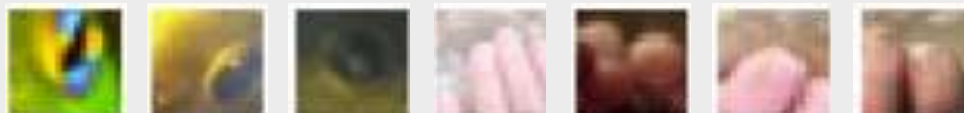
ICLR 2019

Early experiments · architectural test of the texture hypothesis

# CNNs ignore object composition



from tench images



from other images



**Approximating CNNs with bag-of-local-features models works surprisingly well on ImageNet**

W. Brendel, M. Bethge

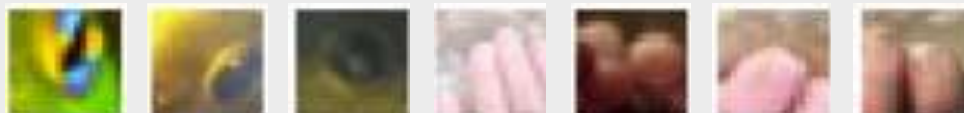
ICLR 2019

Early experiments · architectural test of the texture hypothesis

# CNNs ignore object composition



from tench images



from other images



**Approximating CNNs with bag-of-local-features models works surprisingly well on ImageNet**

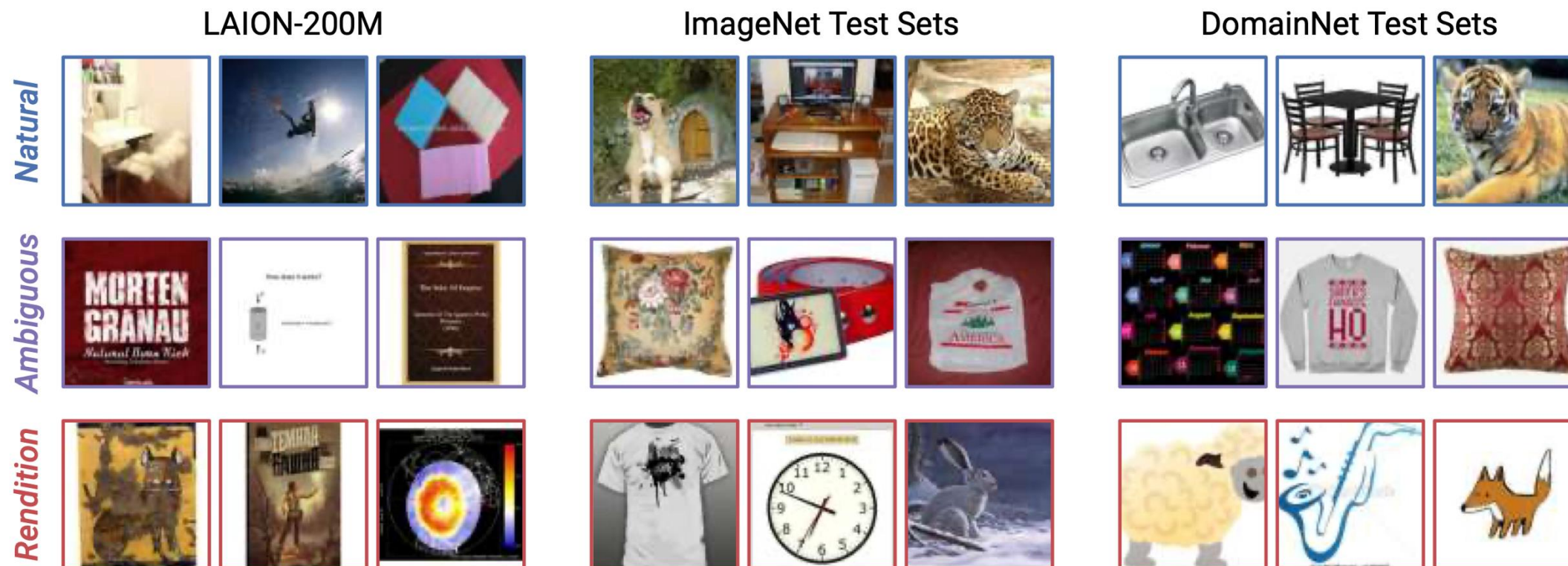
W. Brendel, M. Bethge

ICLR 2019

**Interventional probes  
make competing  
explanations disagree**

Foundation models changed the experiment

# What does generalization mean in the era of foundation models?



**In Search of Forgotten Domain Generalization**

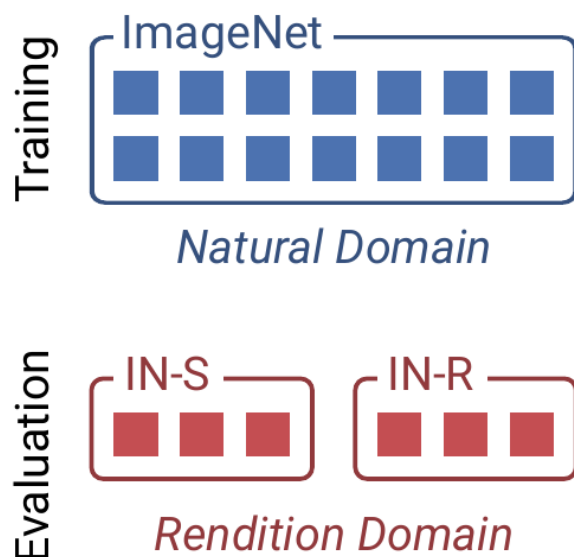
P. Mayilvahanan, R. S. Zimmermann, T. Wiedemer, E. Rusak, A. Juhos, M. Bethge, W. Brendel  
ICLR 2025 (spotlight)

## Part I

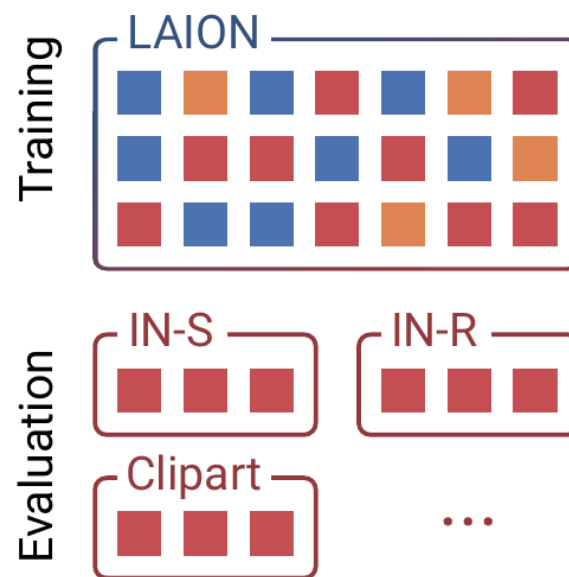
**Does scale create robustness  
to unseen domains?**

# Create a boundary in the training experience

## A ImageNet Era



## B Web-Scale Era



### In Search of Forgotten Domain Generalization

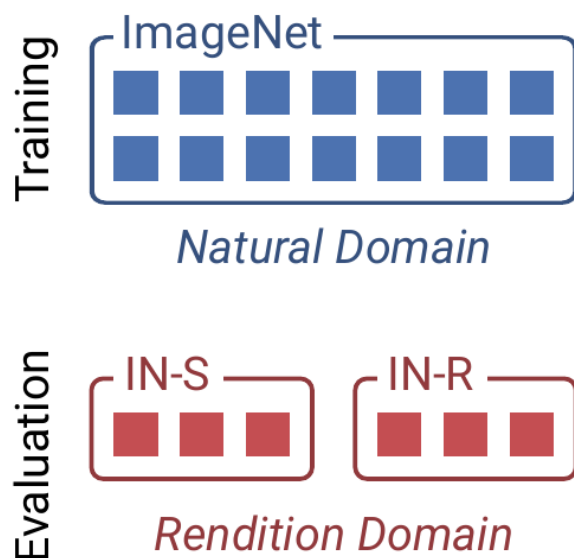
P. Mayilvahanan, R. S. Zimmermann et al. · ICLR 2025 (spotlight)

### Does CLIP's Generalization Performance Mainly Stem from High Train-Test Similarity?

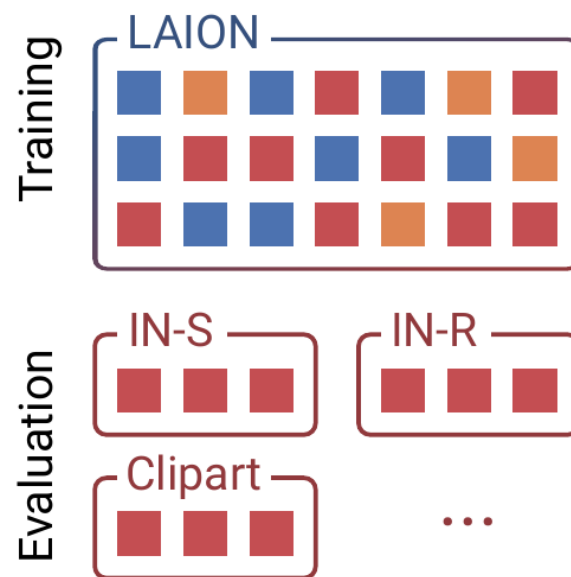
P. Mayilvahanan, T. Wiedemer et al. · ICLR 2024

# Create a boundary in the training experience

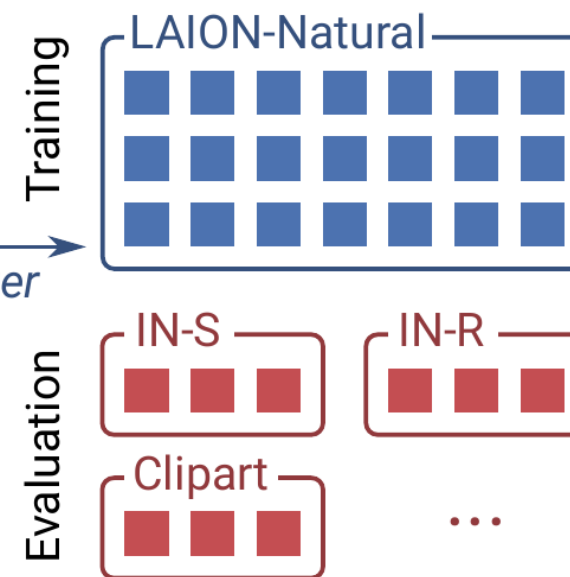
## A ImageNet Era



## B Web-Scale Era



## C Web-Scale + Filter



**In Search of Forgotten Domain Generalization**

P. Mayilvahanan, R. S. Zimmermann et al. · ICLR 2025 (spotlight)

**Does CLIP's Generalization Performance Mainly Stem from High Train-Test Similarity?**

P. Mayilvahanan, T. Wiedemer et al. · ICLR 2024

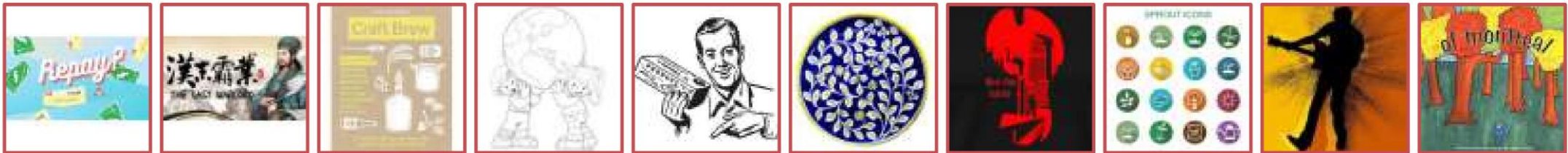
CLIP · an exposure intervention

# Create a boundary in the training experience

LAION-Natural ~57 million samples



LAION-Rendition ~16 million samples



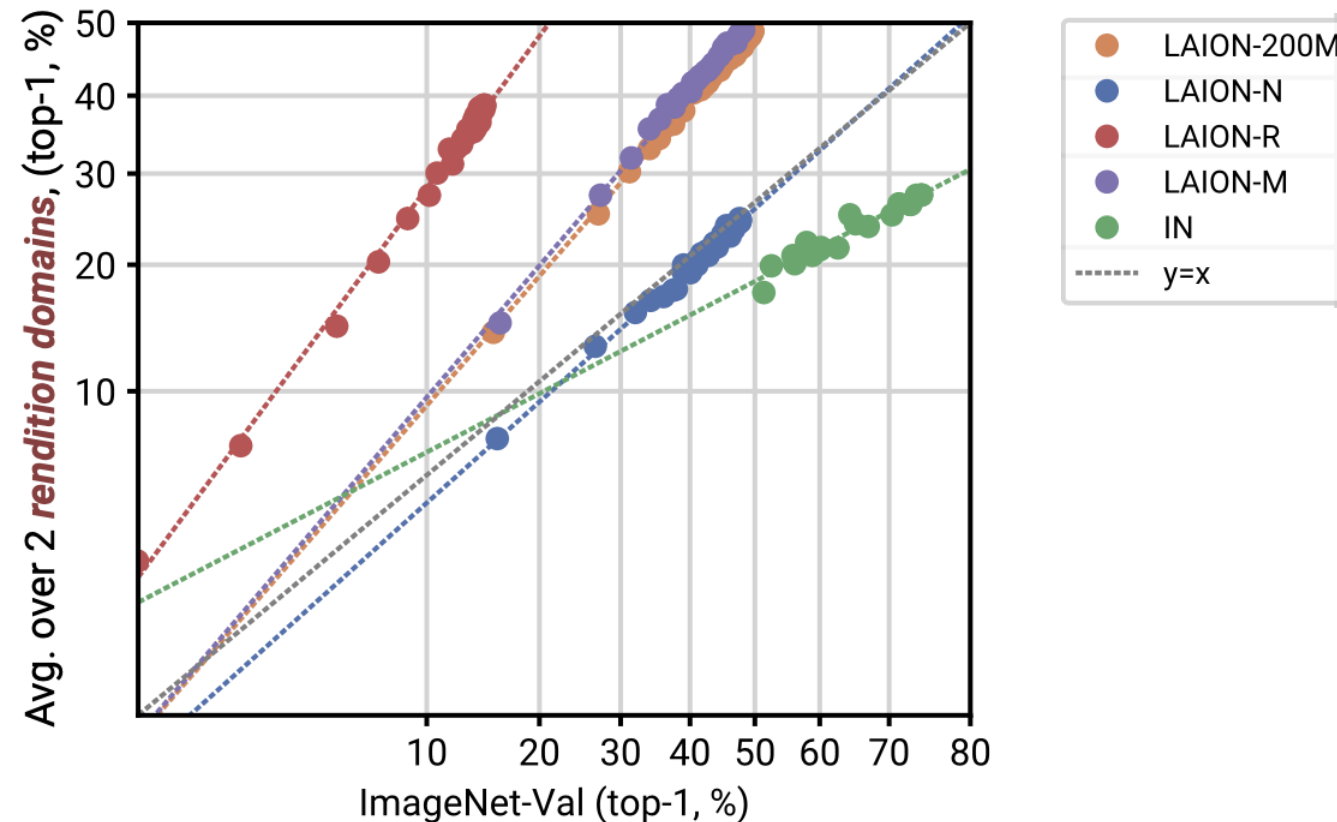
**In Search of Forgotten Domain Generalization**

P. Mayilvahanan, R. S. Zimmermann et al. · ICLR 2025 (spotlight)

**Does CLIP's Generalization Performance Mainly Stem from High Train-Test Similarity?**

P. Mayilvahanan, T. Wiedemer et al. · ICLR 2024

# There is no strong emergence of robustness



## In Search of Forgotten Domain Generalization

P. Mayilvahanan, R. S. Zimmermann et al. · ICLR 2025 (spotlight)

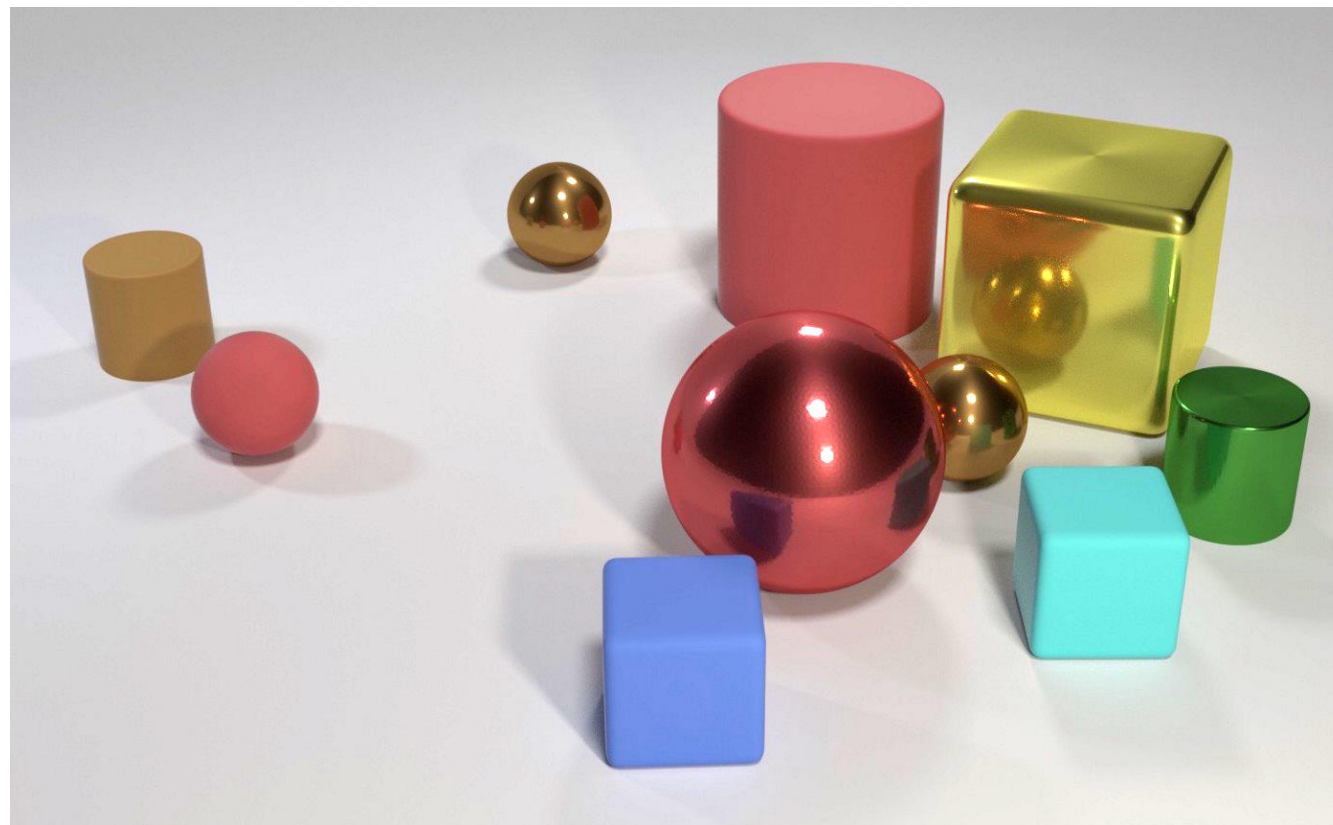
## Does CLIP's Generalization Performance Mainly Stem from High Train-Test Similarity?

P. Mayilvahanan, T. Wiedemer et al. · ICLR 2024

## Part II

# How can neural networks learn object-centric representations?

# Scenes composed of multiple objects



**CLEVR: A Diagnostic Dataset for Compositional Language and Elementary Visual Reasoning**

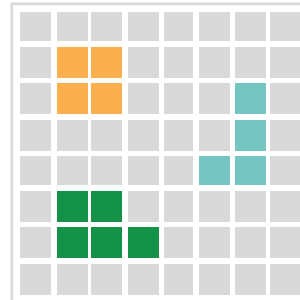
J. Johnson, B. Hariharan, L. van der Maaten, L. Fei-Fei, C. L. Zitnick, R. Girshick

CVPR 2017



# Under what conditions can we isolate object representations?

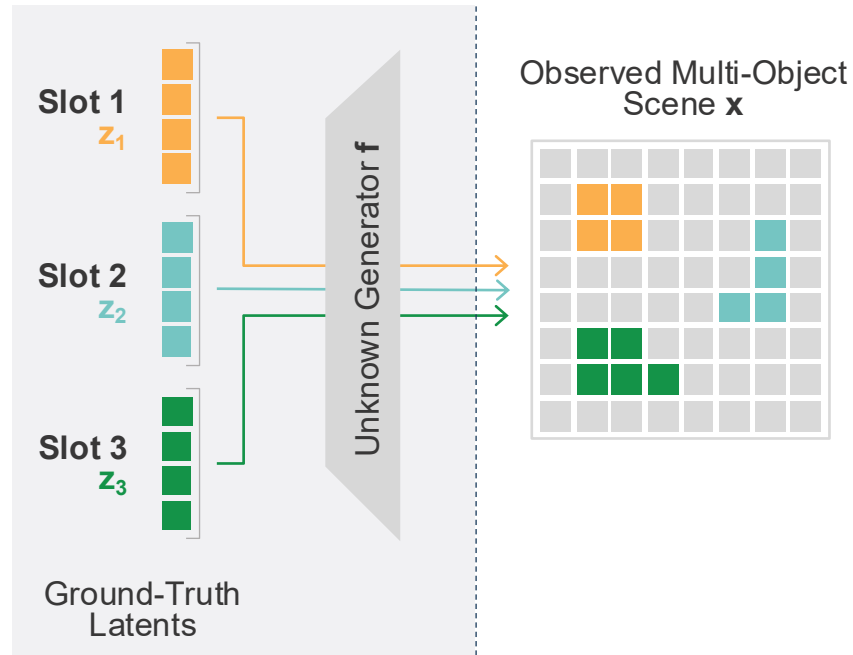
Observed Multi-Object  
Scene  $\mathbf{x}$



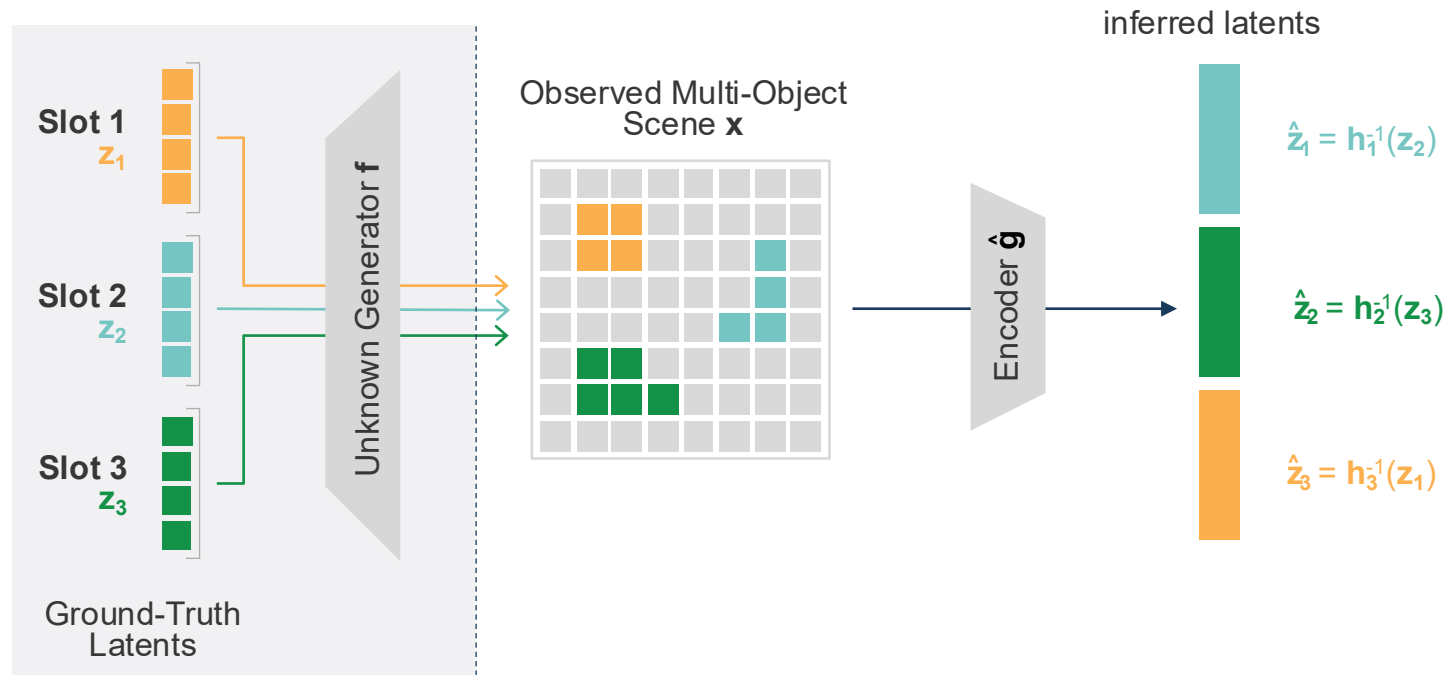
## Provably Learning Object-Centric Representations

J. Brady, R. S. Zimmermann, Y. Sharma, B. Schölkopf, J. von Kügelgen, W. Brendel  
ICML 2023

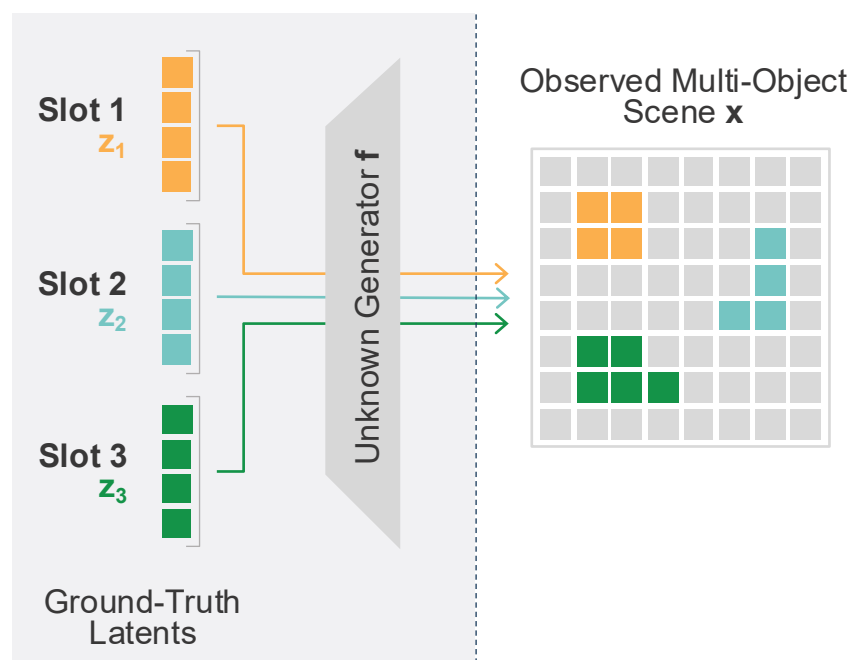
# Under what conditions can we isolate object representations?



# Under what conditions can we isolate object representations?



# We can isolate object representations if the generator has two properties: **compositionality** & **irreducibility**



### # Compositionality

= every pixel belongs to at most one slot.

### # Irreducibility

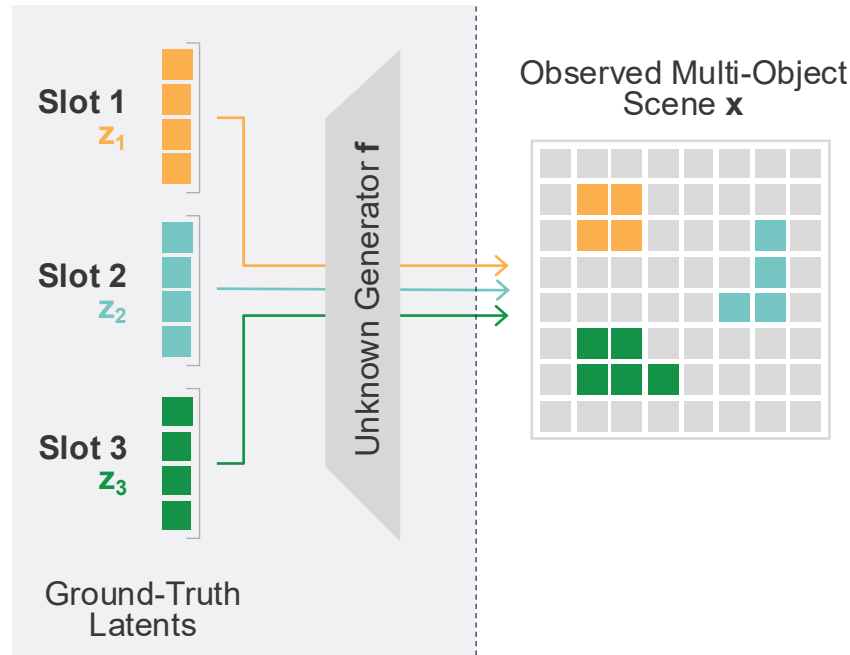
High redundancy across pixels of the same object.



**Provably Learning Object-Centric Representations**

J. Brady, R. S. Zimmermann, Y. Sharma, B. Schölkopf, J. von Kügelgen, W. Brendel  
ICML 2023

We can isolate object representations if the generator has two properties: **compositionality** & **irreducibility**

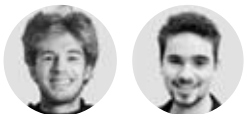
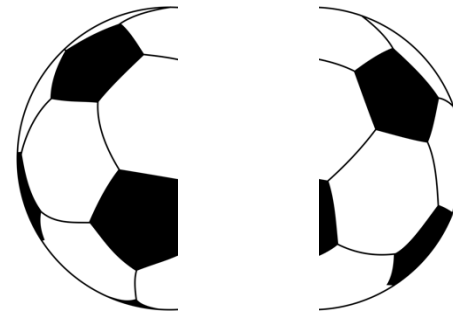


### # Compositionality

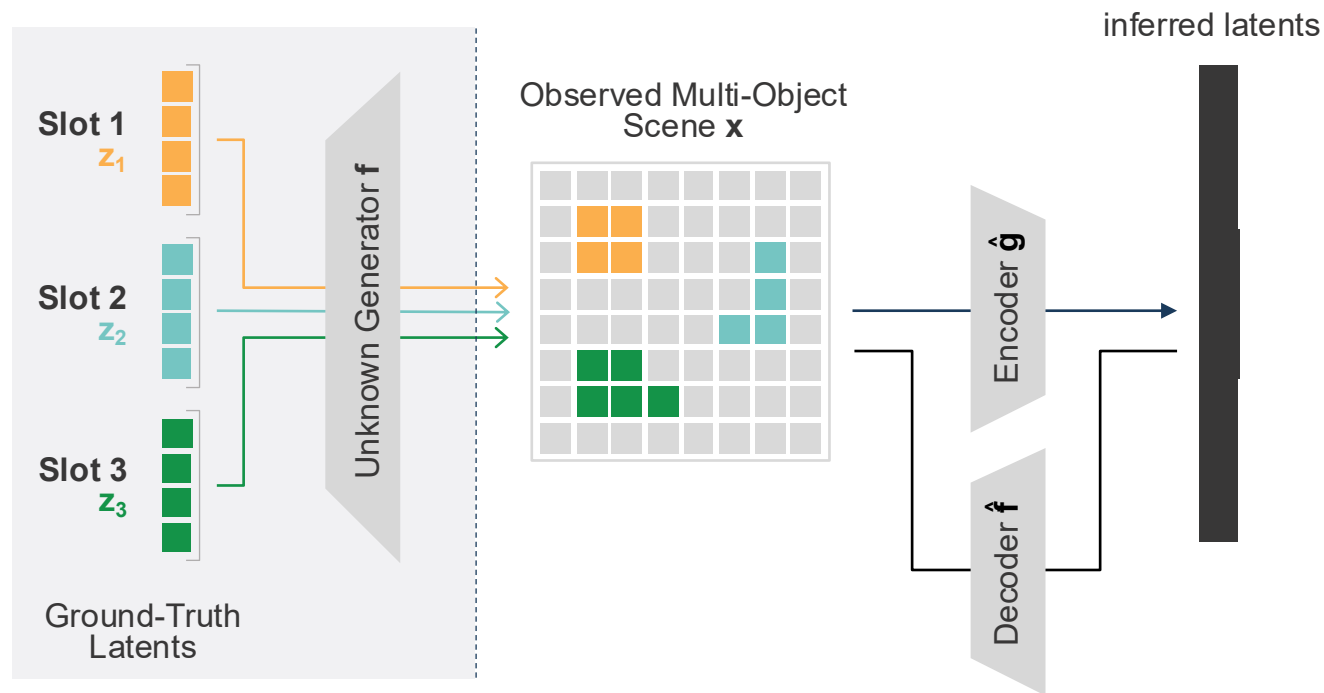
= every pixel belongs to at most one slot.

### # Irreducibility

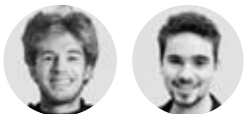
High redundancy across pixels of the same object.



# Object representations can be learnt with a simple autoencoder + regularization



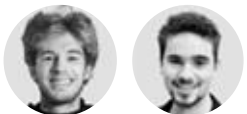
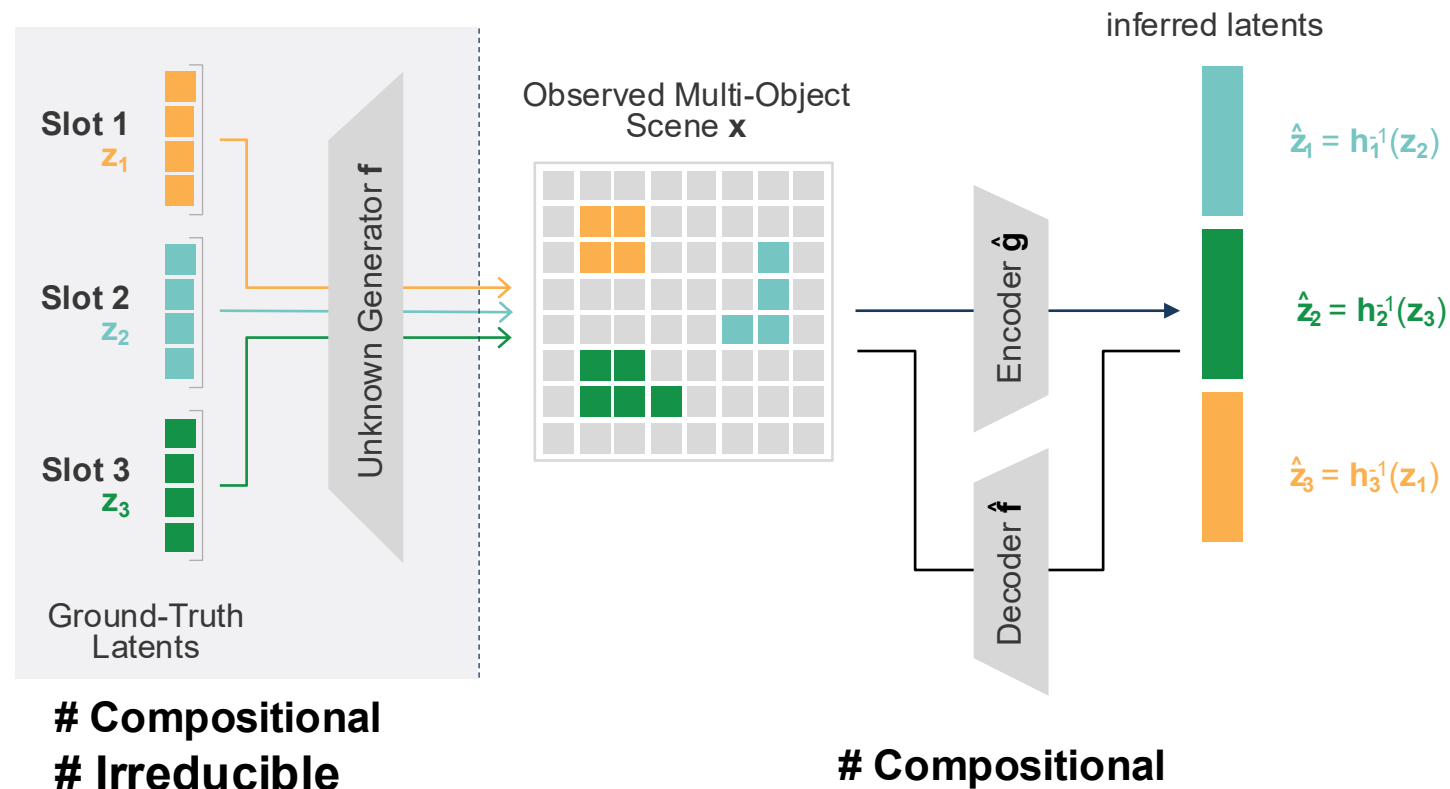
**# Compositional**  
**# Irreducible**



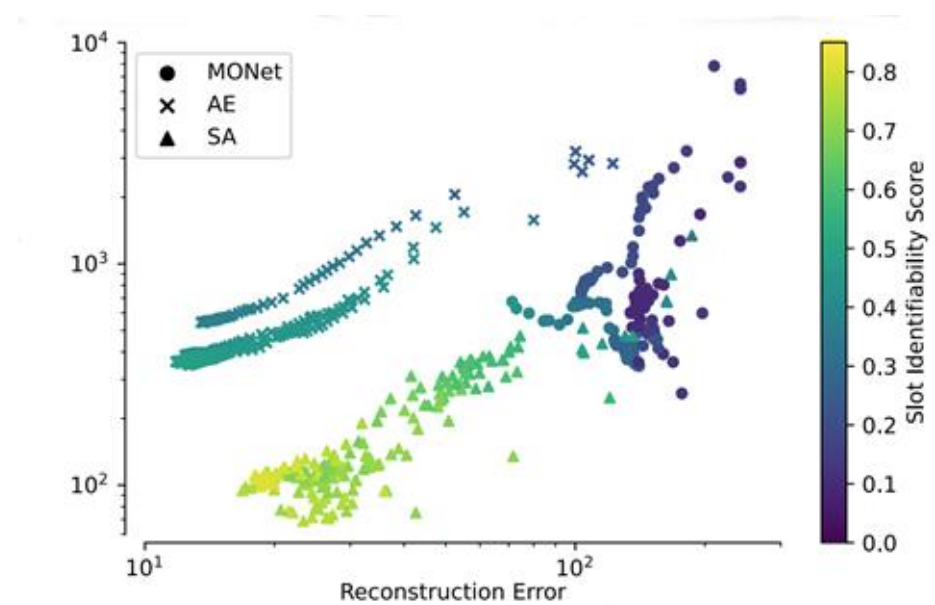
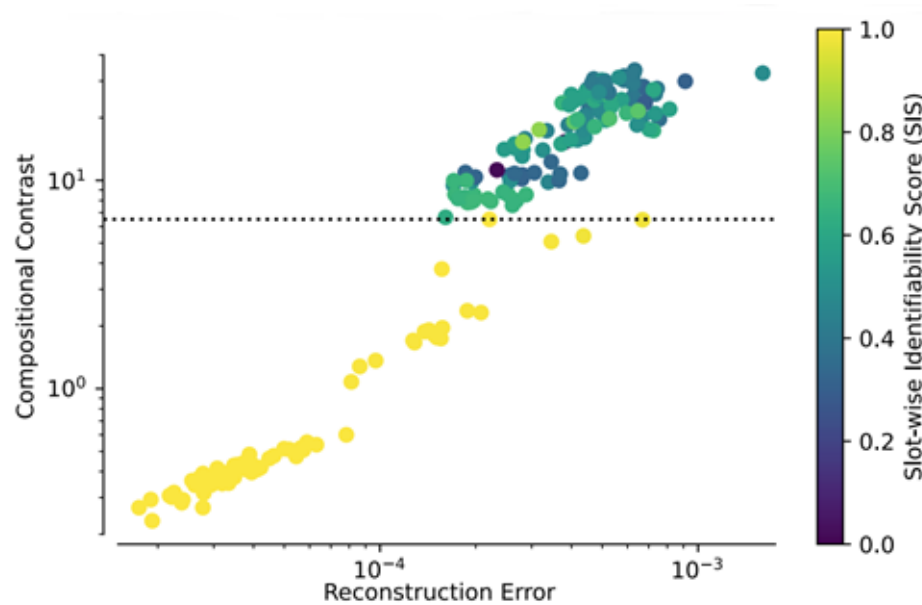
**Provably Learning Object-Centric Representations**

J. Brady, R. S. Zimmermann, Y. Sharma, B. Schölkopf, J. von Kügelgen, W. Brendel  
ICML 2023

# Zero reconstruction + compositional decoder = decomposed representation



# Zero reconstruction + compositional decoder = decomposed representation

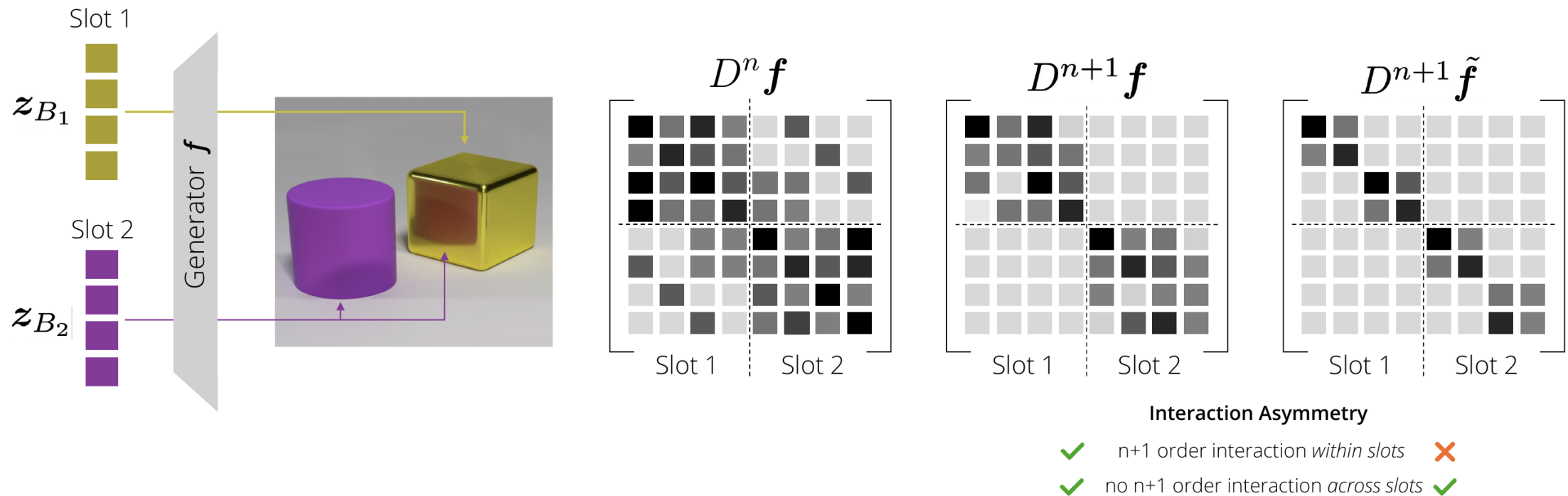


## Provably Learning Object-Centric Representations

J. Brady, R. S. Zimmermann, Y. Sharma, B. Schölkopf, J. von Kügelgen, W. Brendel  
ICML 2023

## Result II

# A unifying principle for learning compositional abstractions



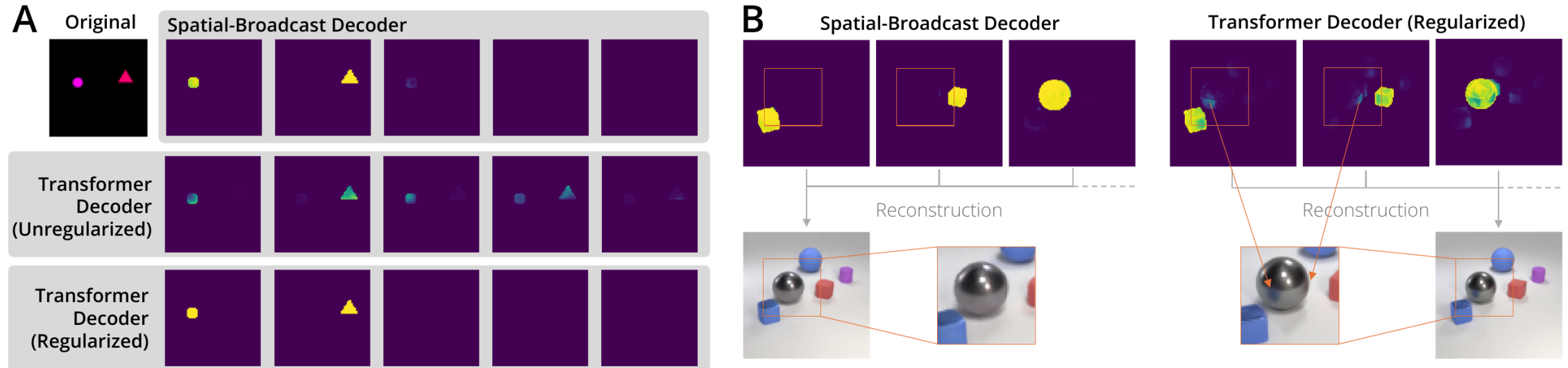
Interaction Asymmetry: A General Principle for Learning Composable Abstractions

J. Brady, J. von Kügelgen, S. Lachapelle, S. Buchholz, T. Kipf, W. Brendel

ICLR 2025

## Result II

# A unifying principle for learning compositional abstractions



**Interaction Asymmetry: A General Principle for Learning Composable Abstractions**

J. Brady, J. von Kügelgen, S. Lachapelle, S. Buchholz, T. Kipf, W. Brendel

ICLR 2025

## Part III

**Do LLMs extract the algorithmic rules that underlie the data?**

# Which rule did the sequence model learn?

**ab    aabb    aaabbb**

**Equal counts of a and b**

**All a's before all b's**

Every training string satisfies both regularities



**Position: Understanding LLMs Requires More Than Statistical Generalization**

P. Reizinger, S. Ujváry, A. Mészáros, A. Kerekes, W. Brendel, F. Huszár

ICML 2024

# An unfamiliar prompt separates the rules

A prompt outside the training grammar

**abaa ...**

**+ bb**

3 a's, 3 b's  
Equal counts restored

**+ bbbb**

3 a's, 5 b's  
Counts stay unequal

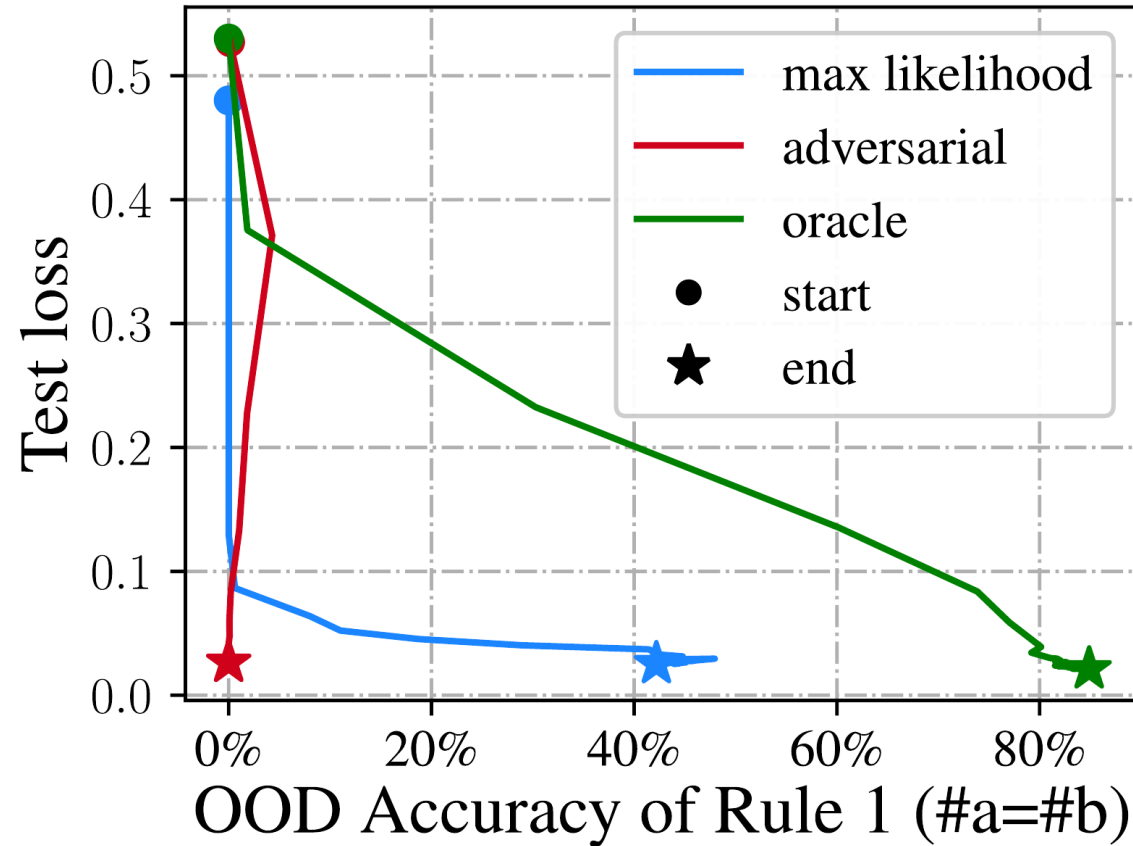


**Position: Understanding LLMs Requires More Than Statistical Generalization**

P. Reizinger, S. Ujváry, A. Mészáros, A. Kerekes, W. Brendel, F. Huszár

ICML 2024

# Similar test loss, different rule generalisation



**Position: Understanding LLMs Requires More Than Statistical Generalization**

P. Reizinger, S. Ujváry, A. Mészáros, A. Kerekes, W. Brendel, F. Huszár

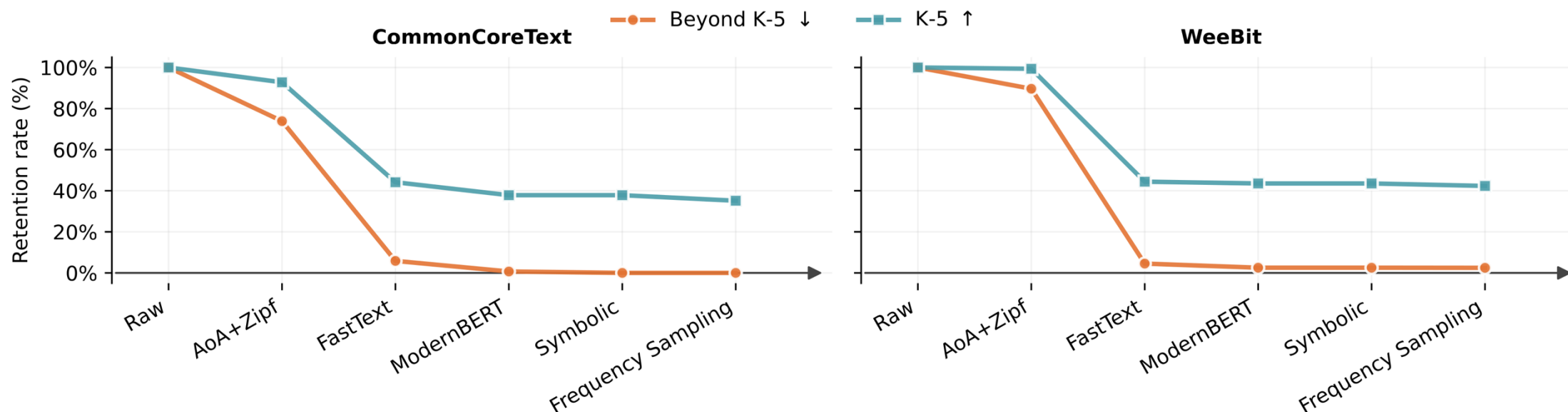
ICML 2024

## Part IV

# How can we cleanly probe learning in LLMs?

# Control the curriculum before testing generalisation

Same 5B model and training recipe. Different curricula.

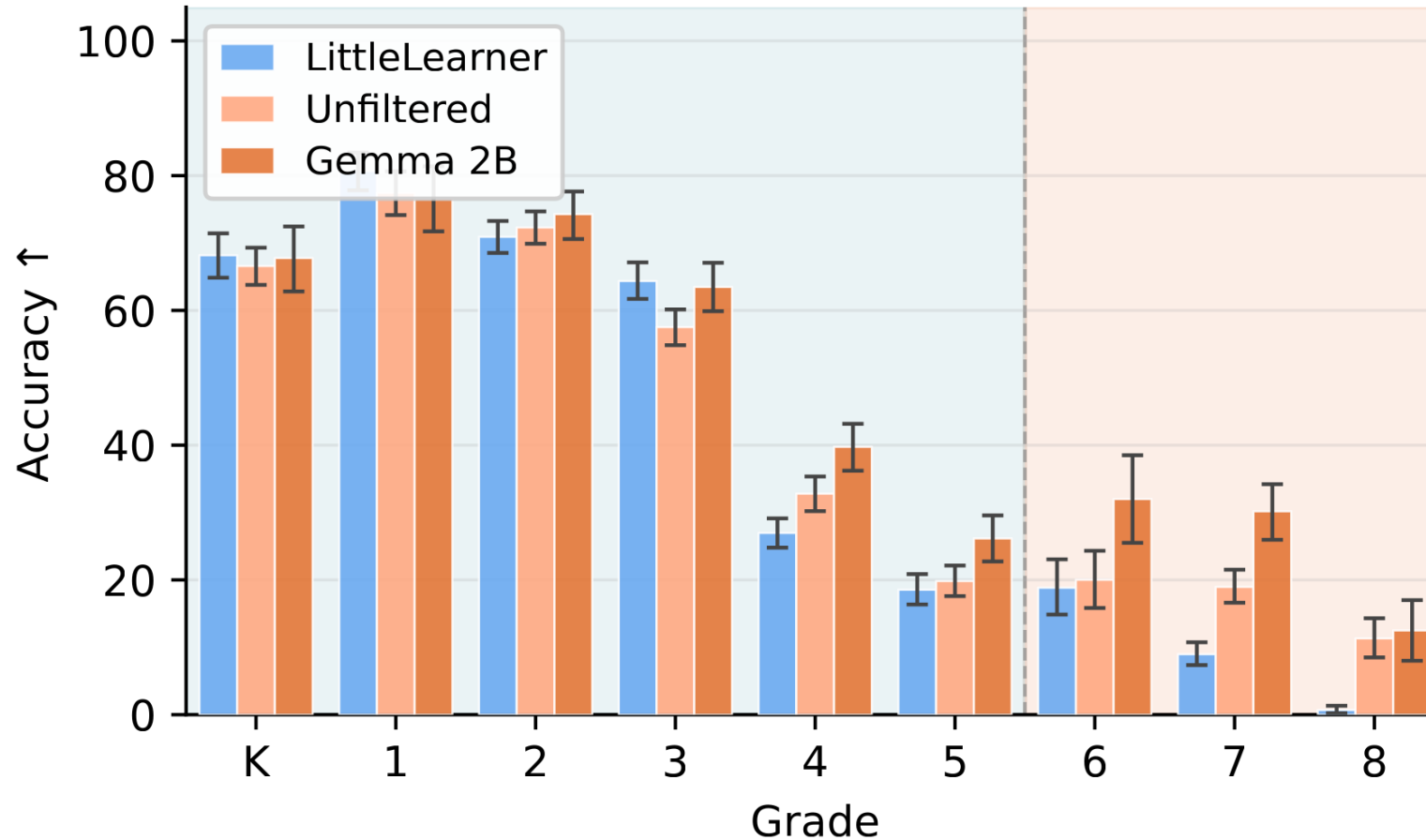


**LittleLearner: Language Models Under Pedagogically Controlled Knowledge Exposure**

F. Li, J. Zeller, M. Prada-Corral, T. Wiedemer, P. Mayilvahanan, R. Cotterell, W. Brendel

arXiv:2608.13545, 2026

# Performance diverges beyond the intended boundary

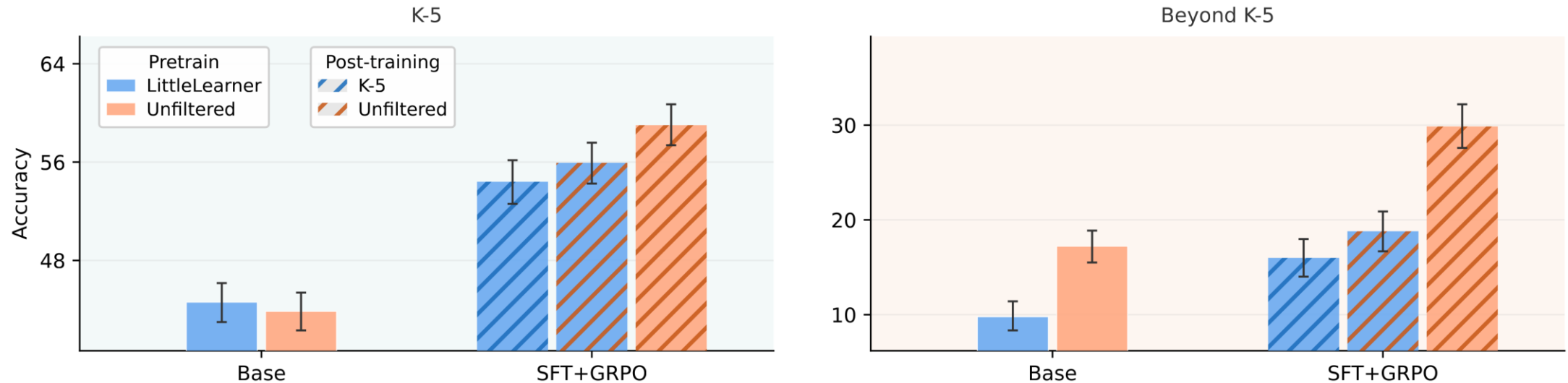


**LittleLearner: Language Models Under Pedagogically Controlled Knowledge Exposure**

F. Li, J. Zeller, M. Prada-Corral, T. Wiedemer, P. Mayilvahanan, R. Cotterell, W. Brendel

arXiv:2608.13545, 2026

# What does further training change?



**Further learning helps; prior exposure still matters**



**LittleLearner: Language Models Under Pedagogically Controlled Knowledge Exposure**

F. Li, J. Zeller, M. Prada-Corral, T. Wiedemer, P. Mayilvahanan, R. Cotterell, W. Brendel

arXiv:2608.13545, 2026

# A platform for studying what a model can learn next

## Curriculum learning

Which prerequisites enable transfer?

## Continual learning

What survives the next learning stage?

## Knowledge acquisition

Which experience creates a new skill?



**LittleLearner: Language Models Under Pedagogically Controlled Knowledge Exposure**

F. Li, J. Zeller, M. Prada-Corral, T. Wiedemer, P. Mayilvahanan, R. Cotterell, W. Brendel

arXiv:2608.13545, 2026

**Targeted generalisation interventions  
can support or reject theories  
of representation learning**



Jack Brady  
*PhD*



Thomas Klein  
*PhD*



Fanfei Li  
*PhD*



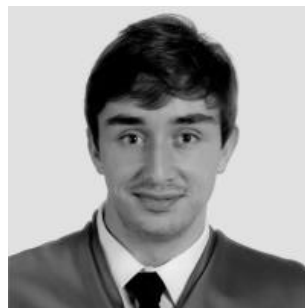
Attila Juhos  
*PhD*



Patrik Reizinger  
*PhD*



Jana Zeller  
*PhD*



Manuel de Prada  
*PhD*



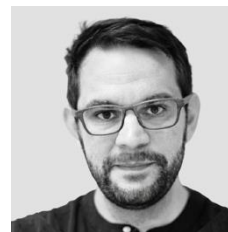
Abhinav Menon  
*PhD*



Matthias Bethge  
*Uni Tübingen*



Nicolas Carlini  
*Google Brain*



Ferenc Huszár  
*Cambridge*



Justin Gilmer  
*Google Brain*



Bernhard Schölkopf  
*MPI-IS, Tübingen*



Oliver Bringmann  
*Uni Tübingen*



Aleksander Madry  
*MIT*



Felix Wichmann  
*Uni Tübingen*





**Ontic Labs**

# Build physical intelligence with us

Foundation models that learn how the physical world works

Research Scientist  
Head of Engineering  
Research Intern

**Open roles and applications**



[onticlabs.io](https://onticlabs.io)